CariSurg MedTech Pathways 2026
Week 6 Baseline Model Report

Student: Ashi Baldie

Date: 14 July 2026

Programme: CariSurg MedTech Pathways 2026

GitHub Repository

https://github.com/baldiea75-tech/carisurg-portfolio.git

Summary

Two baseline classifiers, a logistic regression model and a decision tree, were trained to predict Emergency Severity Index (ESI) triage level from a patient's vital signs and presenting complaint. Both were compared against a stratified random baseline, a model that guesses at random while respecting the same class balance as the training data, so any real model has a fair floor to beat.

Logistic regression is the stronger baseline and is recommended for Week 7. It correctly identifies one in four of the most critical patients, where the decision tree identifies none at all. Neither model is yet safe to trust without a clinician reviewing every case, and the reasons why are set out below.

Results

1.1 Setup

The dataset used is the 55,121 patient Yale EMMLC dataset assessed in Week 5. The target is esi , the nurse assigned triage level from 1, most critical, to 5, least critical. The two leakage columns identified in Week 5, disposition and previousdispo, are excluded entirely, since both are only known after a triage decision has already been made. Demographic and administrative columns are also excluded, so both models work only from vital signs and presenting complaint flags, the information available the moment a patient arrives.

The data was split 80 percent training and 20 percent test, stratified on esi , meaning both the training set and the test set were made to keep the same mix of triage levels rather than being split purely at random. This matters because ESI 1 is rare enough that an ordinary random split could leave the test set with almost no ESI 1 patients to evaluate on at all. A fixed random seed of 42 was used throughout, for the split itself, for the logistic regression model and for the decision tree, and is committed to the project README so the split is reproducible by any reviewer re-running the notebook.

The decision tree's maximum depth was capped at 5 rather than left unbounded. This keeps the tree shallow enough to be drawn as a short flowchart and explained to a clinician in a single sitting, at some cost to raw accuracy. An unbounded tree would likely score higher on paper while becoming both unexplainable and prone to memorising the training data rather than learning patterns that generalise.

1.2 Overall Benchmark

Before reading these figures, three terms are worth defining plainly, since they recur throughout this report.

Precision asks: of the patients the model labelled as a given ESI level, how many actually were that level.

Recall asks the reverse question: of the patients who truly were a given ESI level, how many did the model correctly catch. F1 score combines the two into a single number, useful for ranking models but not a substitute for looking at precision and recall separately, since a model can trade one against the other.

Sixteen patients in the test set were truly ESI 1. Logistic regression correctly flagged 4 of them. The decision tree flagged none.

Model	Accuracy	Macro F1	Weighted F1	Recall (ESI 1)
Stratified random baseline	0.375	0.204	0.375	0
Logistic Regression	0.667	0.492	0.661	0.25
Decision Tree	0.556	0.216	0.463	0

1.3 Per-Class Breakdown

Overall accuracy hides how a model performs on individual ESI levels, particularly the rare ones, so both models are broken down by class below.

Logistic regression, per class:

ESI Level	Precision	Recall	F1 Score	Support
1	0.444	0.25	0.32	16
2	0.716	0.608	0.658	3,585
3	0.66	0.758	0.706	5,402
4	0.609	0.587	0.598	1,779
5	0.482	0.111	0.181	243

Decision tree, per class:

ESI Level	Precision	Recall	F1 Score	Support
1	0	0	0	16
2	0.796	0.265	0.398	3,585
3	0.527	0.959	0.68	5,402
4	0	0	0	1,779
5	0	0	0	243

Support is the number of true patients at that ESI level in the test set. The decision tree scores exactly zero across the board for ESI levels 1, 4 and 5, meaning it never once predicts any of these three levels. Its 0.556 overall accuracy comes entirely from the two largest classes, ESI 2 and ESI 3, while it is structurally incapable of identifying the most critical patients.

1.4 Confusion Matrices

Both confusion matrices are committed to docs/figures/ . Rows are the true ESI level and columns are the predicted ESI level, using the actual clinical values 1 to 5 rather than raw array positions. The diagonal is every correct prediction; everything off the diagonal is a mistake, and where those mistakes land matters more than the headline accuracy figure.

For logistic regression (w6_confusion_logreg.png), most of the missed ESI 1 patients are predicted as ESI 2 rather than something clinically implausible. For the decision tree (w6_confusion_tree.png), every ESI 1, 4 and 5 patient is redirected into either ESI 2 or ESI 3, visibly confirming the collapse described above.

1.5 Macro F1 versus Weighted F1, and Why They Diverge

Macro F1 treats every ESI level as equally important and simply averages the F1 score across all five levels without regard to how many patients are in each. Weighted F1 instead averages the F1 scores in proportion to how many patients are actually in each class, so the two large classes, ESI 2 and ESI 3, dominate the result. This is precisely why the two numbers pull apart on an imbalanced target like this one. The decision tree's weighted F1 of 0.463 looks only moderately worse than logistic regression's 0.661, because both models are competent at the common classes and weighted F1 is dominated by those common classes. Its macro F1 of 0.216, however, is barely above the random baseline's 0.204, because macro F1 gives ESI 1, 4 and 5 the same voice as ESI 2 and ESI 3, and the tree scores zero on all three. Reporting only weighted F1, or accuracy, would have hidden the tree's complete failure on the rare classes behind a deceptively reasonable looking number.

Metric Justification

Chosen primary metric: recall for ESI 1.

ESI 1 is the highest urgency triage level, reserved for patients who need resuscitation. It makes up roughly 0.1 percent of all visits in this dataset, so a model can post a respectable overall accuracy figure while never once correctly identifying a single ESI 1 patient, exactly as the decision tree demonstrates above. The stratified random baseline already reaches 37.5 percent accuracy purely by guessing in line with the training data's class balance; a model only becomes clinically useful once it demonstrably beats that floor on the class that matters most, not on the classes that are easiest to get right because they are common.

Recall for ESI 1 answers one direct clinical question: of the patients who truly needed resuscitation, how many did the model actually flag. Missing an ESI 1 patient, calling someone who needs resuscitation non-urgent instead, is the most serious failure mode a triage support tool can produce, since it delays care for the patient with the least time to spare. Over-flagging a less urgent patient as more urgent than they are costs a clinician a few extra minutes of attention but causes no direct harm to that patient. That asymmetry between the two kinds of mistake, a missed critical patient versus an unnecessary double check, is why recall on ESI 1 specifically, rather than accuracy or precision, is the number this baseline should be judged on.

Failure Mode Reflection

Which patients does the model miss, and why?

The decision tree fails completely on every rare class. With only 61 ESI 1 patients in the entire training set, a tree bounded to depth 5 does not have enough examples to justify carving out a dedicated branch for such a rare outcome, so it collapses toward the two largest classes, ESI 2 and ESI 3, and looks reasonable on overall accuracy while being structurally blind to the sickest patients, and also to ESI 4 and ESI 5 at the other end of the scale.

Logistic regression does meaningfully better but is still missing three out of every four ESI 1 patients. Most of its missed ESI 1 cases are predicted as ESI 2 rather than something clinically implausible, which suggests the model is not confused about the direction of risk, it is simply under-weighting how urgent these particular presentations are relative to slightly less severe ones nearby.

What would this mean for a patient?

For roughly three in four of the most critical patients arriving at Mercer General, this baseline would not raise a flag above the level a moderately urgent patient receives. Concretely, a patient who needs immediate resuscitation could be triaged by the model as merely emergent rather than the most urgent category, delaying the level of attention they need. As a second opinion sitting alongside a clinician who reviews every patient regardless, this baseline is reasonable to trial. As a decision maker operating without human review, it is not yet safe.

Clinical Explainer Video

A one minute video explaining why recall on ESI 1 matters using an ESI 1 example throughout is linked below:

Video link: carisurg-portfolio

Reproducibility Notes

- Random seed: 42, used for the train and test split, logistic regression and the decision tree. This seed is also stated in the project README.
- Train and test split: 80 percent train, 20 percent test, stratified on `esi` .
- Decision tree depth: capped at 5, chosen for explainability over raw accuracy, as set out in 1.1.
- Dataset: `yaleemmlc_admissionprediction_triage.csv` , unchanged since the Week 5 profiling.
- Confusion matrix figures committed to `docs/figures/w6_confusion_logreg.png` and `docs/figures/w6_confusion_tree.png` .

Full code, both confusion matrices, and the complete per-class breakdown are in `notebooks/Week6_Baseline_Models.ipynb` .